PADRI VJEKO CENTRE

L5-NIT Class

Module: GENPP501 - Python Programming

Learning Outcome 1: Prepare Python Environment

Learning Hours: 15

Indicative Content

1. Selection of Python Tools

- **Python Programming Overview**
  - **Definition:** Python is a high-level, interpreted, general-purpose programming language.
  - **Benefits:**
    - Easy to learn and use (simple syntax).
    - Cross-platform compatibility.
    - **Large community support:** Millions of developers and learners contribute, share resources, and solve problems.

    > **Why this matters:** Help is always available through forums, tutorials, and meetups.

  - **Rich set of libraries and frameworks:** Pre-built tools for tasks or entire applications.
    - **Libraries:** math, pandas, matplotlib
    - **Frameworks:** Django, Flask, TensorFlow
- **Characteristics of Python**
  - **1. Interpreted and Dynamically Typed**
    - Python executes code line by line and identifies variable types automatically.

    ```
    x = 5       # integer
    x = "five"  # now it's a string
    ```

    > Magic box analogy: it adapts to whatever you put in.

  - **2. Object-Oriented (OOP)**
    - Code structured around classes (blueprints) and objects (instances).

    ```
    class Car:
        def __init__(self, brand):
            self.brand = brand

    my_car = Car("Toyota")
    print(my_car.brand)
    ```

    > Cookie cutter analogy: class = cutter, object = cookie.

  - **3. Extensible and Embeddable**
    - Integrate C/C++ code or embed Python in other programs.

    > Smartphone analogy: add apps or embed inside another system.

  - **4. Readability and Simplicity**
    - Easy to read, English-like syntax, indentation instead of braces.

    ```
    for i in range(3):
        print("Hello")
    ```

    > Like reading a simple recipe step by step.

- **Applications of Python**
  - **1. Data Science** – Machine Learning, AI, Statistics

    > Example: Predicting house prices from past data.

    > Like teaching a student with examples.

  - **2. Software Development** – Web Apps, Desktop Apps, APIs

    > Example: Instagram backend partly uses Python.

Like Lego blocks – build small pieces, combine into full app.

- **3. Automation** – Scripting, Task Automation, DevOps

  Example: Script downloads daily reports automatically.

  Like a personal assistant handling repetitive tasks.

- **4. Data Analytics** – Data Processing, Visualization, Reporting

  Example: Company analyzes sales data for decision-making.

  Turning raw ingredients (data) into a cooked meal (insights).

- **Identification of Python Tools**
  - Python Interpreter (CPython, PyPy, Jython)
  - IDEs (PyCharm, VS Code, Jupyter Notebook, Spyder)
  - Package Managers (pip, conda)
  - Virtual Environments (venv, virtualenv, conda envs)

2. Installation of Python Tools

- **A. Computer System Requirements**
  - **Hardware:** Processor ≥1GHz, RAM ≥2GB (4GB recommended), Storage ≥200MB
  - **Software:** Windows, Linux, macOS; updated OS with admin rights
- **B. Install Python**
  - Download from Python.org
  - Optionally, use Anaconda for scientific computing (includes libraries)
- **C. Configure Virtual Environment**
  - Create:

    ```
    python -m venv env
    ```

  - Activate:

    ```
    Windows: .\env\Scripts\activate
    Linux/macOS: source env/bin/activate
    ```

  - Install packages:

    ```
    pip install requests
    pip install pandas
    ```

  Virtual environment = separate school bag for each project/subject.

3. Testing Python Installation

- **1. Run Python Version**

  ```
  python --version
  or python3 --version
  ```

  Output: Python 3.12.5

- **2. Check Python Interpreter**

  ```
  python
  or python3
  ```

  Output: >>> prompt ready for commands

- **3. Test Package Manager (pip)**

  ```
  pip --version
  pip install requests
  ```

  Output: pip 24.0 from ... (Python 3.12)

  Checking version = engine, interpreter = starting car, pip = adding parts